

DDR基础课程

第三节 - 初步了解DRAM Jedec Spec

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目录

关注的问题

- 初步了解DRAM Jedec Spec
- bank architecture是什么？什么是ba, bg？以及什么是rank？什么是channel？
- 端口信号有哪些？
- 命令有哪些？为什么需要这些命令？
- 命令与状态的关系是什么？
- 命令的格式是什么样的？
- 命令与数据的波形是什么样的？

Bank Architecture

- Bank Group用于弥补存储单元速度较低的缺陷
- 可以通过增加Burst Length来弥补这个缺陷
- 但是增加Burst Length会降低灵活性，即最小的数据访问单元会增加

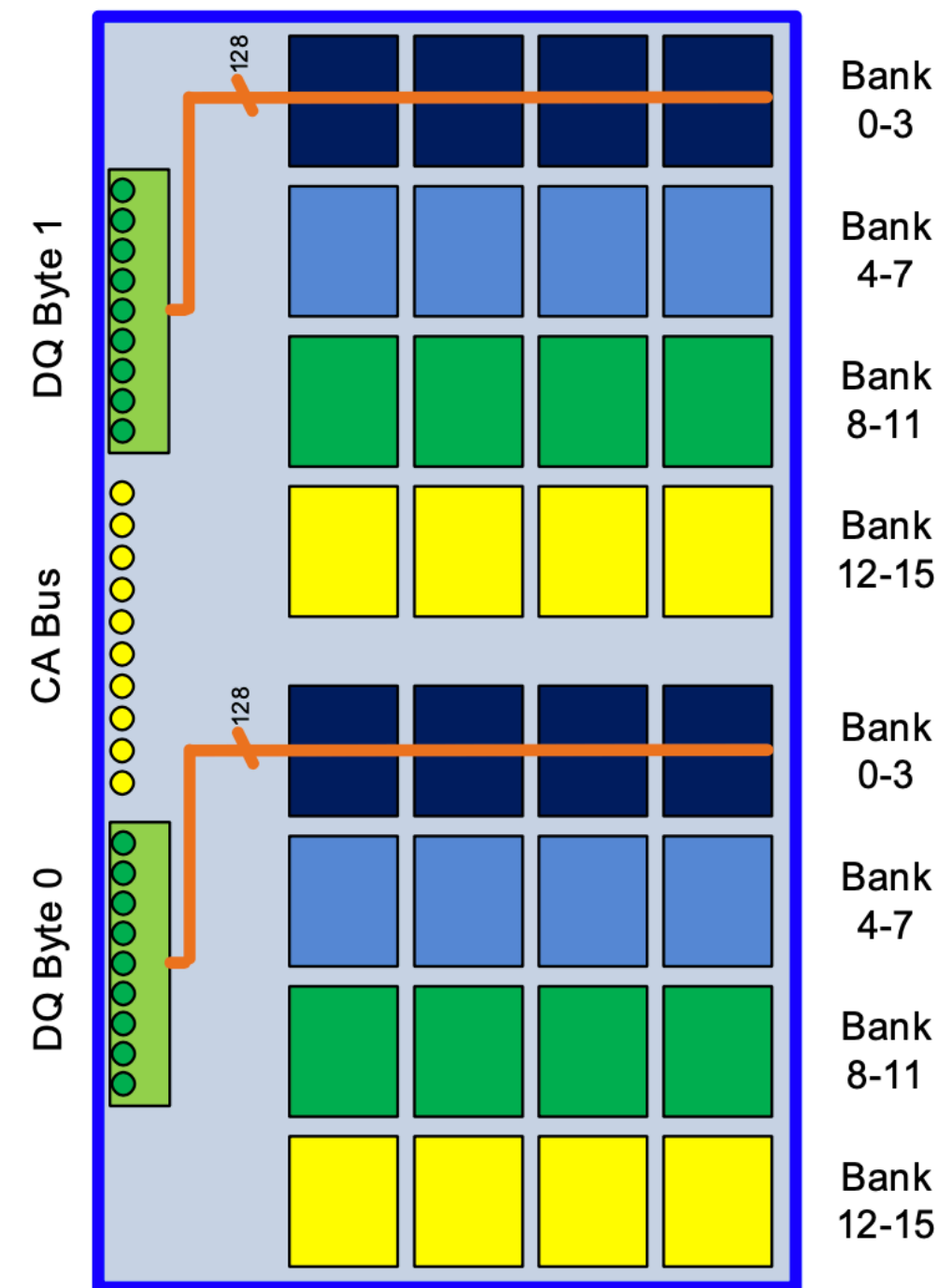


Figure 3 — 16B Mode Configuration Example (One Channel Shown)

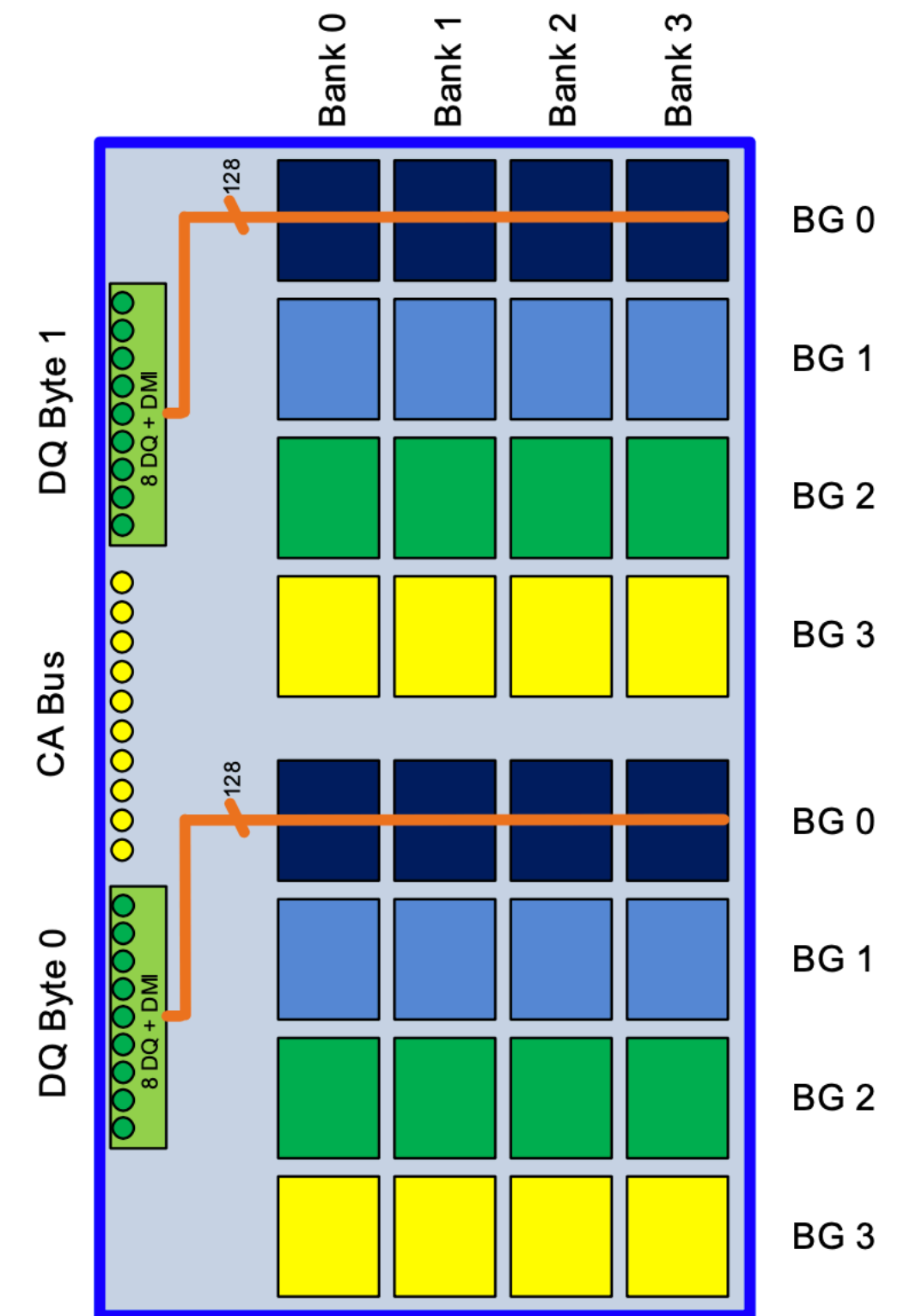
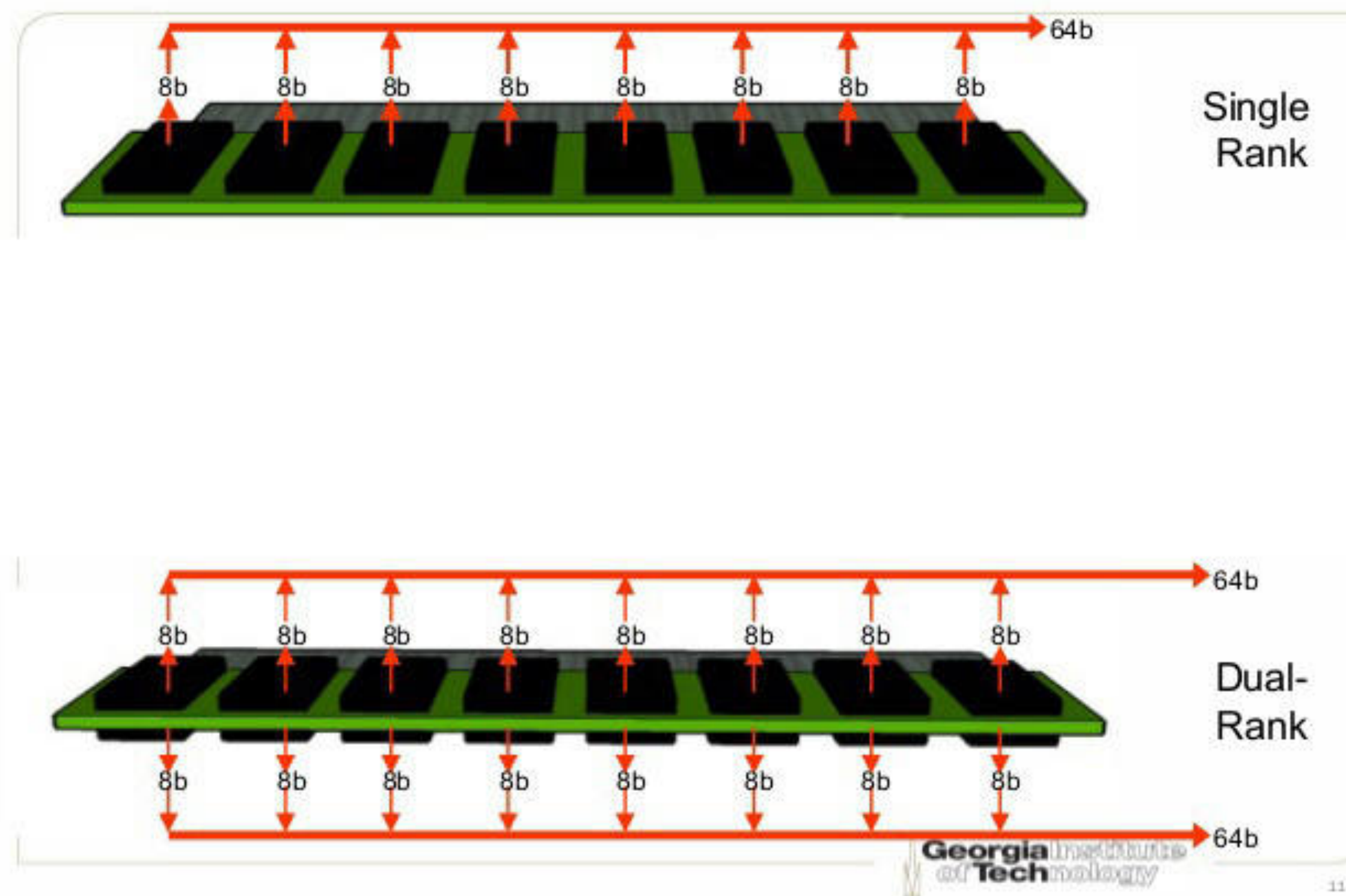


Figure 1 — BG Mode Configuration Example (One Channel Shown)

rank是什么?

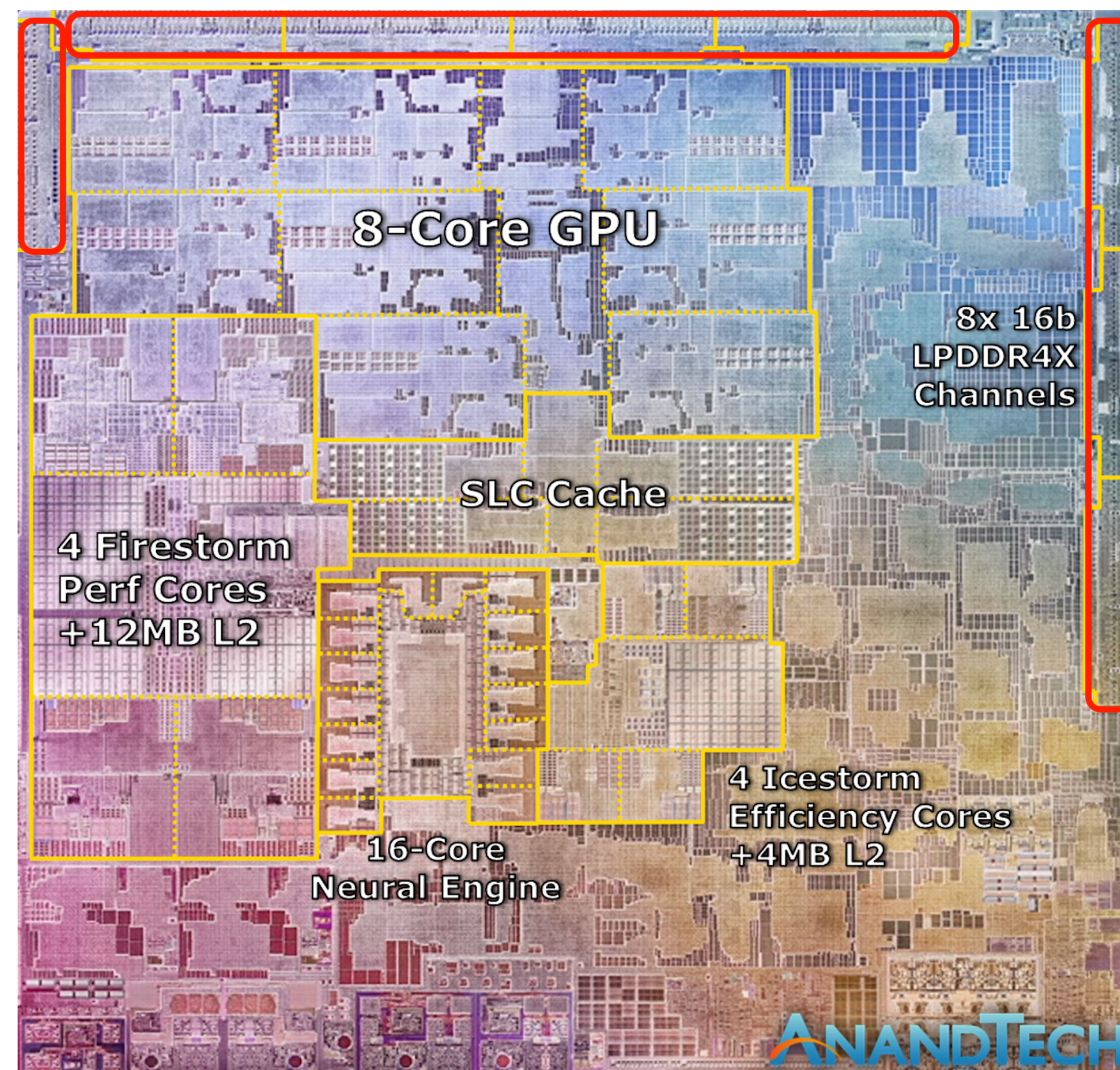
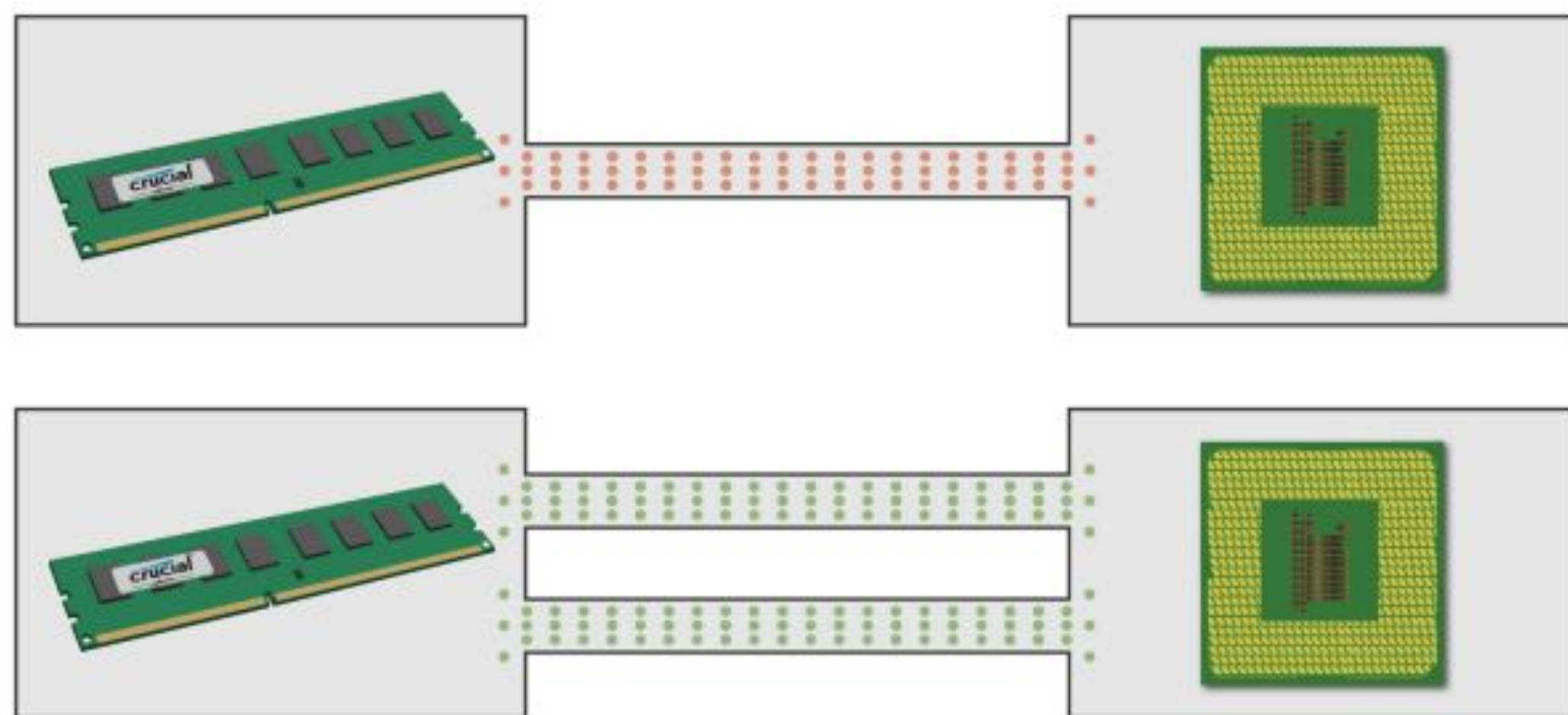
- 目的是为了在同一个通道上增加容量
- cs不共用
- ca及dq信号共用
- 使用的时候会在看两个Rank间切换

DRAM Ranks



channel是什么?

- 会同时使用多个通道
- DDR X64为一个通道
- LPDDR X16为一个通道



为什么要多个通道?

- 问题1: 系统需要5GB/s以上的带宽, 现在我们采取的是LPDDR5X, 一个Channel的模式, 问: 需要跑多高的速率?
- 问题2: 系统需要30GB/s以上的带宽, 现在我们采用的是LPDDR5X - 8533Mbps的模式, 问: 一个Channel可以满足吗?

- 分析1:
- 公式: $\text{bandwidth} = \text{data rate per pin} * \text{number of pins}(\text{data strobe width})$
- $5\text{GB/s} = 5 * 8 \text{ Gb/s} = \text{data rate} * 16(1 \text{ Channel})$
- $\text{data rate} = 2.5\text{Gbps} = 2500\text{Mbps}$
- 我们知道LPDDR5X的最高速率是8533Mbps, 因此这个速率是可以满足的

为什么要多个通道？

- 问题1: 系统需要5GB/s以上的带宽，现在我们采取的是LPDDR5X，一个Channel的模式，问：需要跑多高的速率？
- 问题2: 系统需要30GB/s以上的带宽，现在我们采用的是LPDDR5X - 8533Mbps的模式，问：一个Channel可以满足吗？

- 分析2:
- 公式: $\text{bandwidth} = \text{data rate per pin} * \text{number of pins}(\text{data strobe width})$
- $30\text{GB/s} = 30 * 8 \text{ Gb/s} = 8533\text{Mbps} * 16 * N \text{ Channel}$
- $N \text{ Channel} = 1.76$
- 因此一个Channel满足不了这样的带宽需求，需要两个才能满足

端口信号

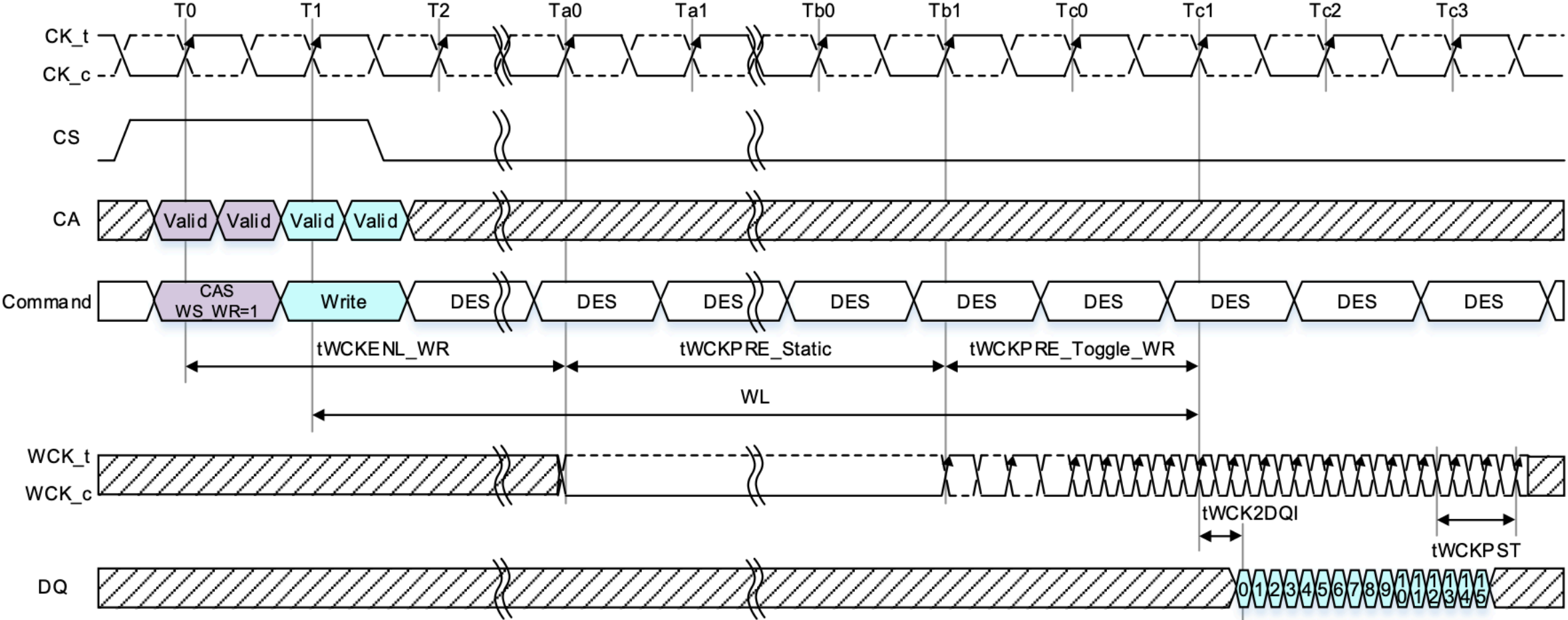


Table 1 — Pad Definition and Description

Symbol	Type	Description	Note
CK_t, CK_c	Input	Clock: CK_t and CK_c are differential clock inputs. All Double Data Rate (DDR) Command/Address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising(falling) edge of CK_t (CK_c) and second crossing point is falling(rising) edge of CK_t (CK_c). Single Data Rate (SDR) inputs, CS is sampled on the crossing point that is the rising(falling) edge of CK_t (CK_c).	
CS	Input	Chip Select: CS is part of the command code and is sampled on the rising(falling) edge of CK_t (CK_c) unless the device is in power-down or Deep Sleep mode where it becomes an asynchronous signal.	
CA[6:0]	Input	Command/Address Inputs: CA signals provide the Command and Address input according to the Command Truth Table	
DQ[15:0]	I/O	Data Input/Output: Bi-direction data bus.	
WCK[1:0]_t WCK[1:0]_c	Input	Data Clocks: WCK_t and WCK_c are differential clocks used for WRITE data capture and READ data output.	
RDQS [1:0]_t , RDQS[1:0]_c	RDQS_t :I/O, RDQS_c :Output	Read Data Strobe: RDQS_t and RDQS_c are the differential output clock signals used to strobe data during a READ operation. And RDQS_t is also used as a Parity pin at Write with Link Protection enabled.	1
DMI[1:0]	I/O	Data Mask Inversion: DMI achieves multiple function such as Data Mask (DM), Data Bus Inversion (DBI), and Parity at read with ECC operation by setting the Mode Register and DMI is a bi-directional signal, and each byte of data has a DMI signal.	1
ZQ	Reference	ZQ: ZQ is used to calibrate the output drive strength and the termination resistance as calibration reference. There is one ZQ pad per die. The ZQ pin shall be connected to VDDQ through a 240Ω ±1% resistor.	
VDDQ, VDD1, VDD2H, VDD2L	Supply	Power Supplies: Isolated on the die for improved noise immunity.	
VSS	GND	Ground Reference: Power supply ground reference	
RESET_n	Input	RESET: When asserted LOW, the RESET_n signal resets the die. RESET_n is an asynchronous signal.	
NOTE 1 See Table 2 and Table 3, Per-Byte Signal List/Description, for multi-function definition for these pins.			

课后问题

- 课后问题1: Apple M1芯片的DDR通道数为____? 其对应的DRAM类型为____? 最高速率为____?最高总带宽为____?
- 课后问题2: “如果总带宽不足, 可以通过将Rank数翻倍, 从而将带宽翻倍”。以上说法是正确的吗?
- 课后问题3: “data rate指的是CK的频率的两倍”。以上说法是正确的吗?
- 课后问题4: “data rate指的是WCK的频率”。以上说法是正确的吗?
- 课后问题5: DMI信号应该与以下哪个时钟对齐? A. CK; B. WCK
- 课后问题6: ZQ信号外接的电阻大小为____欧?

下节预告

第四节 - DRAM Jedec Spec从用户的角度看读写

- 命令
- 状态
- 从用户的角度看如何读取一个数据回来
- 从用户的角度看如何写一个数据进去

谢谢

命令

- 真值表

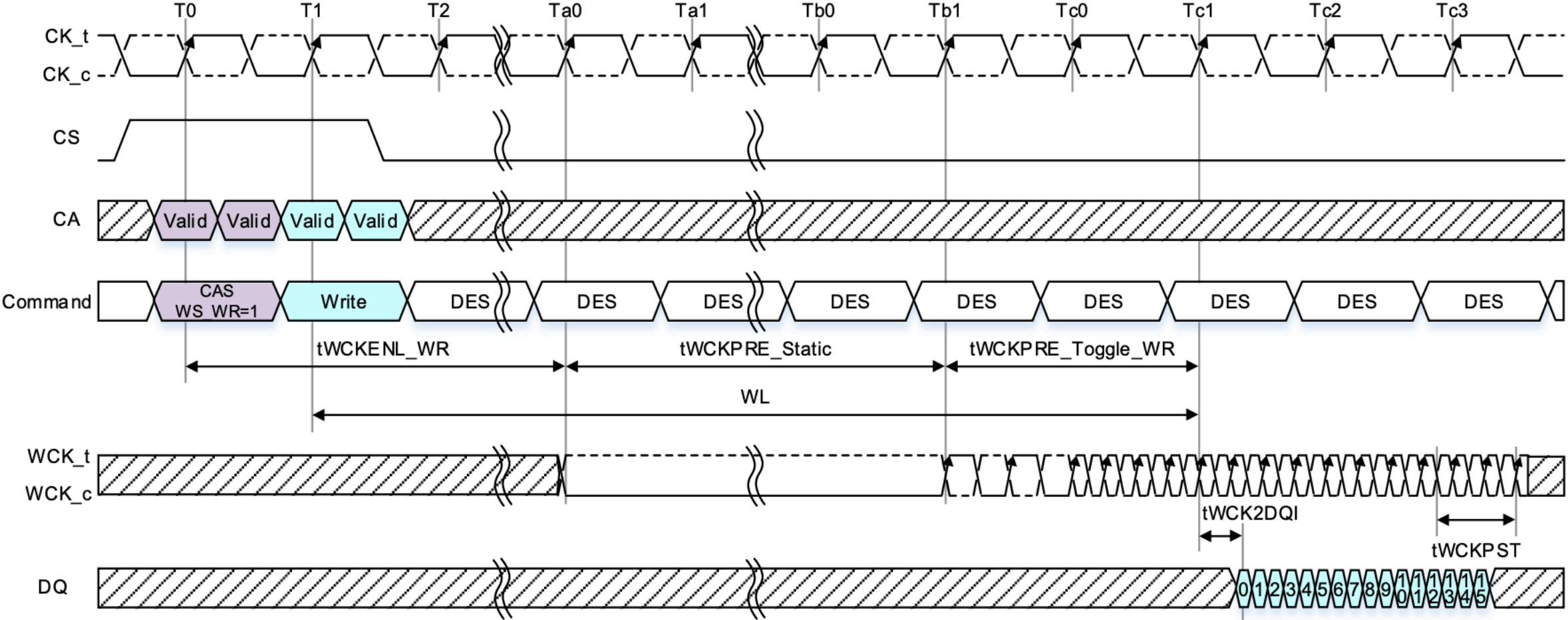


Table 197 — Command Truth Table

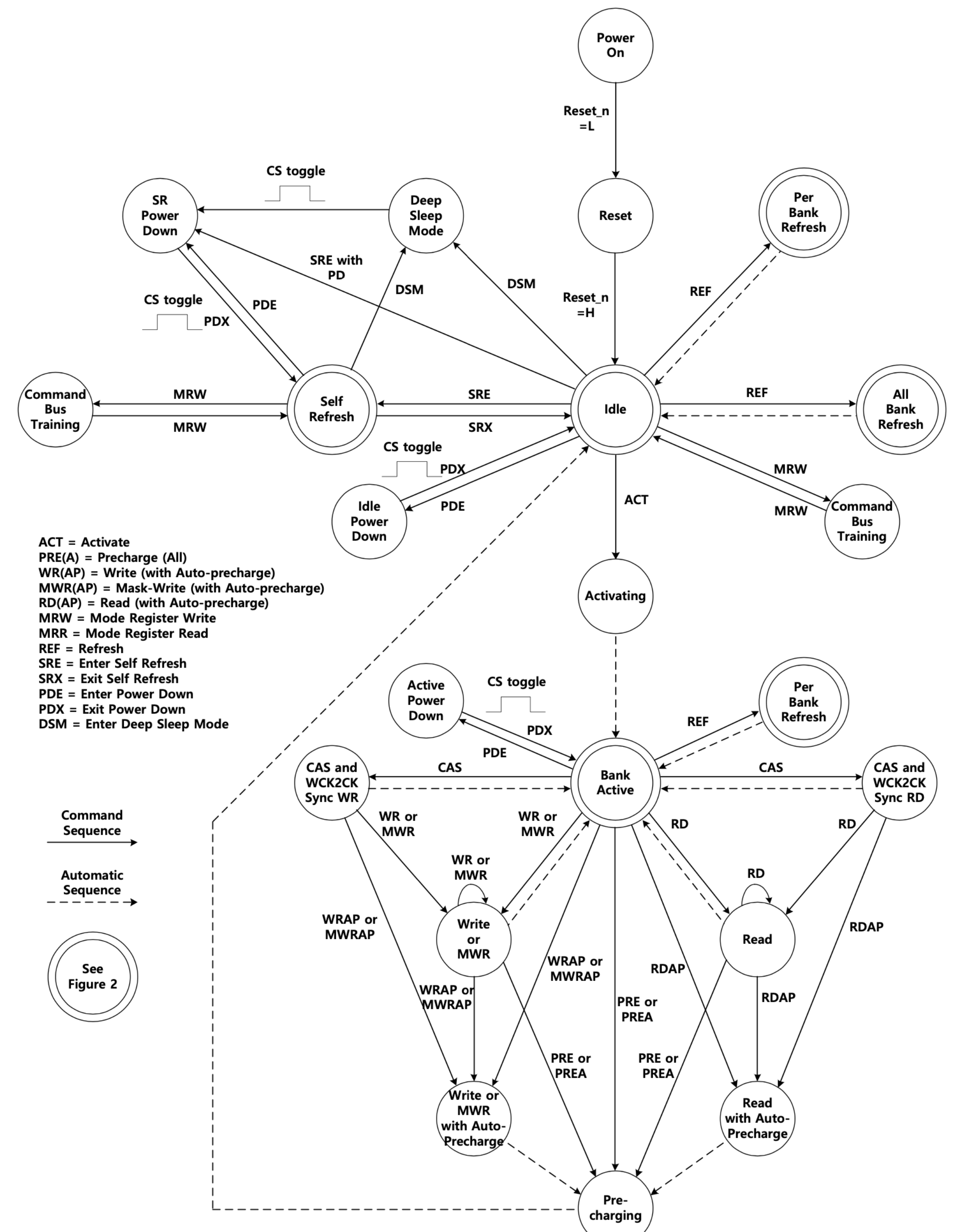
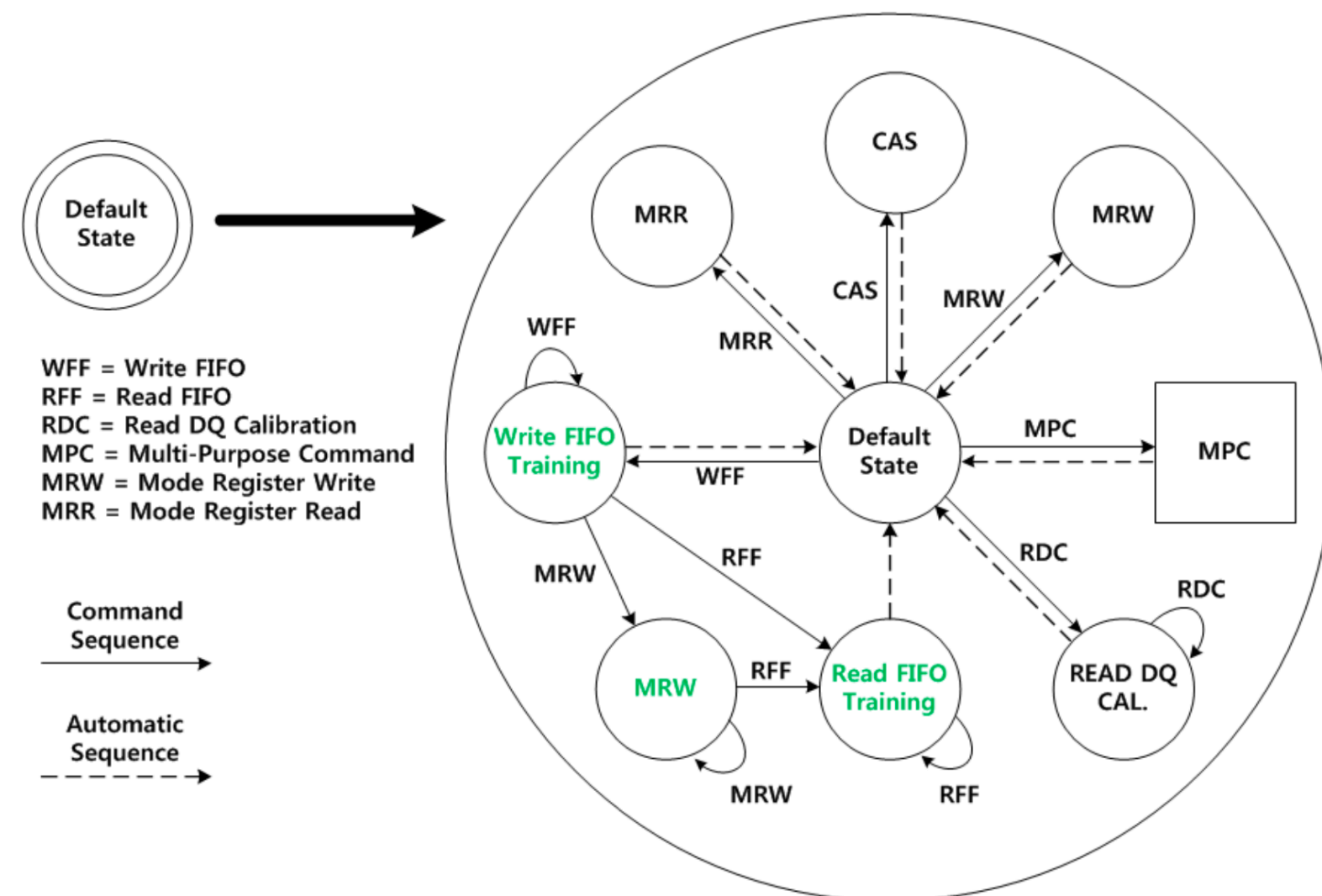
SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
		CS	CA0	CA1	CA2	CA3	CA4	CA5	CA6		
DESELECT (DES)	Any	L	X	X	X	X	X	X	X	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
NO OPERATION (NOP)	Any	H	L	L	L	L	L	L	L	R1	1, 2
		X	X	X	X	X	X	X	X	F1	
POWER DOWN ENTRY (PDE)	Any	H	L	L	L	L	L	L	H	R1	1, 2, 12
		L	X	X	X	X	X	X	X	R2	
ACTIVATE-1 (ACT-1)	Any	H	H	H	H	R14	R15	R16	R17	R1	1, 2, 3, 4
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	R11	R12	R13	F1	
ACTIVATE-2 (ACT-2)	Any										1, 2, 4
			BA0	BA1	BA2	BA3					
		X	R0	R1	R2	R3	R4	R5	R6	F1	
PRECHARGE (PRE) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	H	R1	1, 2, 3, 5
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	V	V	AB	F1	
REFRESH (REF) (Per Bank, All Banks)	Any	H	L	L	L	H	H	H	H	R1	1, 2, 3, 5, 14
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	RFM	SB0	V	F1	
MASK WRITE (MWR)	Any	H	L	H	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
WRITE (WR16 or WR)	Any	H	L	H	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
WRITE32 (WR32)	BG/16B	H	L	L	H	L	C3	C4	C5	R1	1, 2, 3, 6, 7, 8
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
READ (RD16 or RD)	Any	H	H	L	L	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	
READ32 (RD32)	BG/16B	H	H	L	H	C0	C3	C4	C5	R1	1, 2, 3, 6, 7, 10, 11
	BG		BA0	BA1	BG0	BG1					
	16B	X	BA0	BA1	BA2	BA3	C1	C2	AP	F1	

- 真值表

Operation Mode	CAS Command Operand(s)						
	WS_WR	WS_RD	WS_FS	DC0-3	WRX	WXSA	WXSBB/B3
WCK2CK SYNC (WR)	H	L	L	LLLL	L	L	L
WCK2CK SYNC (WR) + Write Data Copy	H	L	L	VVVV	L	L	L
WCK2CK SYNC (WR) + Write X	H	L	L	LLLL	H	V	V
WCK2CK SYNC (RD)	L	H	L	LLLL	L	L	V
WCK2CK FAST SYNC	L	L	H	LLLL	L	L	L
WCK2CK SYNC OFF	H	H	H	LLLL	L	L	L
Write Data Copy	L	L	L	VVVV	L	L	L
Write X	L	L	L	LLLL	H	V	V
Non-Zero Burst Start Address for RD	L	L	L	LLLL	L	L	H
WCK SUSPEND	H	L	H	LLLL	L	L	L
Illegal		Other Operands Combination					
NOTE 1 "V" means "H" or "L" (a defined logic level). NOTE 2 All CAS command operands are non-sticky. NOTE 3 Write Data Copy function requires DC0-3 operands not to be all-zero. NOTE 4 CAS-WCK_SUSPEND command is valid only when MR0 OP[2]=1 _B (Enhanced WCK Always On mode supported) and MR18 OP[4]= 1 _B (WCK Always On mode enabled)							

SDRAM COMMAND	BK ORG (BG, 16B, 8B)	SDR CMD PIN	DDR COMMAND PINS							CK_t edge	Notes
			CS	CA0	CA1	CA2	CA3	CA4	CA5		
CAS	Any	H	L	L	H	H	WS_WR	WS_RD	WS_FS	 R1	1, 2, 7, 13
		X	DC0	DC1	DC2	DC3	WRX	WXSA	WXSB/B3	 F1	
MULTI PURPOSE COMMAND (MPC)	Any	H	L	L	L	L	H	H	OP7	 R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	 F1	
SELF REFRESH ENTRY (SRE)	Any	H	L	L	L	H	L	H	H	 R1	1, 2, 9
		X	V	V	V	V	V	DSM	PD	 F1	
SELF REFRESH EXIT (SRX)	Any	H	L	L	L	H	L	H	L	 R1	1, 2
		X	V	V	V	V	V	V	V	 F1	
MODE REGISTER WRITE-1 (MRW-1)	Any	H	L	L	L	H	H	L	H	 R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	 F1	
MODE REGISTER WRITE-2 (MRW-2)	Any	H	L	L	L	H	L	L	OP7	 R1	1, 2
		X	OP0	OP1	OP2	OP3	OP4	OP5	OP6	 F1	
MODE REGISTER READ (MRR)	Any	H	L	L	L	H	H	L	L	 R1	1, 2
		X	MA0	MA1	MA2	MA3	MA4	MA5	MA6	 F1	
WRITE FIFO (WFF)	Any	H	L	L	L	L	L	H	H	 R1	1, 2
		X	L	L	L	L	L	L	L	 F1	
READ FIFO (RFF)	Any	H	L	L	L	L	L	H	L	 R1	1, 2
		X	L	L	L	L	L	L	L	 F1	
READ DQ CALIBRATION (RDC)	Any	H	L	L	L	L	H	L	H	 R1	1, 2
		X	L	L	L	L	L	L	L	 F1	
NOTE 1 LPDDR5 commands are one clock cycles long and defined by the states of CS at the rising edge (R1) of clock and CA[6:0] at the rising edge (R1) and the falling edge (F1) of clock. Note that some operations such as ACTIVATE, MODE REGISTER WRITE or WCK2CK SYNC (WS_RD, WS_WR, WS_FS) require two commands to initiate. NOTE 2 "V" means "H" or "L" (a defined logic level). "X" means don't care in which case CK_t, CK_c and CA[6:0] can be floated. NOTE 3 Bank Group Address BG[1:0] and Bank Address BA[1:0] determine which bank is to be operated upon. NOTE 4 An ACTIVATE-1 command must be followed by an ACTIVATE-2 command. Only CAS, WRITE, MASK WRITE, READ, PRECHARGE (to a different bank) commands can be issued between ACTIVATE-1 and ACTIVATE-2 commands. The maximum timing gap between ACTIVATE-1 and ACTIVATE-2 commands are 8 clock cycles. An ACTIVATE-1 command must be issued first before issuing an ACTIVATE-2 command. Once an ACTIVATE-1 command is issued, an ACTIVATE-2 command must be issued before issuing another ACTIVATE-1 command. NOTE 5 AB "HIGH" during PRECHARGE or REFRESH command indicates that command must be applied to all banks and bank address is a don't care.											

状态机



课后问题

- 课后问题1: Apple M1芯片的DDR通道数为___? 其对应的DRAM类型为___? 最高速率为___? 最高总带宽为___?
- 课后问题2: “如果总带宽不足, 可以通过将Rank数翻倍, 从而将带宽翻倍”。以上说法是正确的吗?
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- 课后问题4: “data rate指的是WCK的频率”。以上说法是正确的吗?
- 课后问题5: “DDR指的是数据是双沿的。但是在LPDDR5中的命令也是上下沿采样的”。以上说法是正确的吗?
- 课后问题6: LPDDR5中CAS命令中与WCK2CK Sync有关的命令有___个? 分别为___? 他们的含义分别是___?